Cloth-Grasp Tilt & Depth Experiment Plan

Date: 2026-06-03 **Engineer**: Shiyao Ni **Hardware**: Franka FR3 + arm_client, `fr3_pose_controller` **Contact-force source**: Franka `F_ext` (flange, base frame), bias-corrected per run. A301 FlexiForce calibration currently blocked; F_ext is the first-pass estimate.

Objective

Characterize cloth-grasp contact force and tilt tolerance across four gripper variants (rigid vs soft) to quantify the soft-grasper advantage for thin cloth.

Research questions

1. **Force vs depth** — How does contact force scale with press depth across rigid and soft grippers? (Study 1)
2. **Tilt × depth tolerance** — How does tilt about the contact point affect grasp success and contact force, and does this depend on press depth? (Study 2)

Variables

Independent - Gripper variant (4 levels) - Press depth (gripper-specific levels) - Tilt axis ∈ {x, y, z}, angle ∈ {0, 2.5°, 5°}

Dependent - Contact force `F_ext_z` (Franka external wrench, base frame, bias-corrected) - Grasp success (visual: cloth held through lift A → B) - Settle (steady-hold) force - Peak press force

Controlled - Cloth specimen (single, characterized) - Position A / B in robot base frame - Per-gripper contact-point TCP offset (from touch-off pilot) - Approach / descent / press speeds, settle times - Baseline EEF orientation = (180°, 0°, 0°) RPY

Gripper roster

ID	Gripper	Grasp mode
G1	Parallel jaw, standard (rigid)	force-close, 20 N command
G2	Parallel jaw + TPU 90 A pad	force-close, 20 N command
G3	Finray, 33 mm	fixed gap, 60 mm
G4	Finray, 18 mm	fixed gap, 60 mm

Pilot phase (per gripper, one-off)

#	Task	Method	Output
P1	Touch-off at common (x, y)	`probe_surface.py`, 5 taps @ 2 N	`z_table` (mean ± std), TCP offset Δz

#	Task	Method	Output
P2	Max-depth escalation, tilt = 0	step 1 mm until controller reflex (force-safety / accel-discontinuity)	`max_depth_safe`
P3	Success threshold scan, tilt = 0	step 1 mm from −1 mm upward; ≥ 2/3 trials grasp	`d_min_success`

≈ 30–40 trials total across grippers.

Study 1 — depth sweep, tilt = 0

Headline force-vs-depth curve per gripper.

Gripper	Depths (mm)	Repeats	Trials
G1	−1, 0, 1, 2, 3, 4, 5 (capped by `max_depth_safe`)	3	21
G2	−1, 0, 1, 2, 3, 4, 5 (capped)	3	21
G3	−1, 0, 2, 4, 6, 8 (capped)	3	18
G4	−1, 0, 2, 4, 6, 8 (capped)	3	18

Study 1 total: 78 trials.

Study 2 — tilt × depth

Tilt about the **per-gripper contact point** (TCP offset from P1). Two depth levels per gripper:

- `d_min` = `d_min_success` — marginal-grasp fragility under tilt
- `d_safe` = (`d_min_success` + `max_depth_safe`) / 2 — comfortable-grasp margin under tilt

Axis convention (confirmed 2026-06-04): the gripper-closing direction is **y**; the gripper is mirror-symmetric across **xz**. By choice (operator directive 2026-06-04), all three axes are tested **single-sided at {0, 2.5, 5°}** — x and z by mirror symmetry, y by electing to characterise one tilt direction only (into the cloth body, not off the edge). If the −y side turns out to differ materially, a follow-up bill can extend it.

Per-gripper tilt conditions (one axis at a time):

#	rx (°)	ry (°)	rz (°)	Depth
T1a	0	0	0	`d_min`
T2a	2.5	0	0	`d_min`
T3a	5	0	0	`d_min`
T4a	0	2.5	0	`d_min`
T5a	0	5	0	`d_min`
T6a	0	0	2.5	`d_min`
T7a	0	0	5	`d_min`
T1b–T7b	(same tilts)			`d_safe`

7 conditions × 2 depths × 3 reps × 4 grippers = **168 trials**.

Study 3 — inclined-seat finray sub-study

The mounting fixture between the Franka Hand carriage and the finray blades is a CAD design parameter. Two fixture variants under comparison (finray 33 mm only):

Variant	Seat angle	Notes
G3-S2.5	2.5°	finray 33 mm on inclined seat
G3-S5	5°	finray 33 mm on inclined seat

The standard (flat-seat) **G3** from the main study provides a third 0°-seat reference at no extra cost (data is already collected by Study 1 and Study 2).

Protocol (Option B — targeted)

Per fixture variant: a depth sweep at tilt = 0 plus a tilt sweep on the **y axis only** (the gripping direction; the inclined seat geometrically biases this exact axis, which is why this slice is the publishable test). Single-sided per the Study 2 axis convention.

Block	Levels	Trials per variant
Depth sweep (tilt = 0)	−1, 0, 2, 4, 6, 8 mm × 3 reps	18
y-axis tilt × depth	3 angles (ry = 0, 2.5, 5°) × 2 depths (`d_min`, `d_safe`) × 3 reps	18

36 trials × 2 variants = 72 trials.

Detection note: with one-sided sweeps, the seat-shift hypothesis is detectable only if the bias goes in the +y direction. If the data is consistent with a negative shift, a follow-up bill can extend the sweep to ±5° on the inclined-seat variants.

Hypothesis

The seat tilt is a built-in geometric offset on the asymmetric tilt axis. If it acts as a **passive alignment correction**, the tilt-tolerance curve should shift along that axis by approximately the seat-angle difference (G3-S5 vs G3-S2.5 ⇒ ~2.5° shift). If the curves change shape without shifting, the seat is changing blade compliance, not alignment — a different (also useful) story.

Adding the flat G3 as a 0° reference gives three points (0°, 2.5°, 5° seat) for a linear or saturating fit on tilt-tolerance shift vs seat angle.

Total budget

Block	Trials
Pilot (P1–P3)	~35
Study 1 (depth × tilt 0)	78

Block	Trials
Study 2 (tilt × depth, all axes single-sided)	168
Study 3 (inclined-seat sub-study, single-sided)	72
Total	**≈ 353 trials**

Estimated runtime: 4–6 sessions.

Open items (to confirm with supervisor)

1. ~~**Gripping direction** in EEF frame.~~ **RESOLVED 2026-06-04:** gripping direction = **y**; mirror plane = **xz**; asymmetric tilt axis = rotation about **y**. By operator directive 2026-06-04, all three axes (including y) are tested single-sided at {0, 2.5, 5°}; the −y side is deferred to a follow-up bill if motivated by data.
2. ~~`GRASP_WIDTH_M` for the **finray 18 mm**.~~ **RESOLVED 2026-06-04:** `GRASP_WIDTH_M = 0.060` (same as G3, finray 33 mm).
3. ~~Whether to **empirically validate** the gripper mirror symmetry on the two symmetric axes (x and z) for G1 (full ±5° on those axes adds 12 trials, provides a "verified line for the report).~~ **DECLINED 2026-06-04:** x and z sweep at {0, 2.5, 5°} only, on the theoretical symmetry argument; the y-axis already covers both signs in the main matrix. If a reviewer pushes back, the validation can be added as a targeted 12-trial follow-up at that point.

All open items resolved or declined. Matrix locked at **≈ 425 trials**.

Recorded per trial (auto-saved)

- pose + F_ext CSV at 30 Hz
- `run_config.json` (gripper, depth, tilt, positions, controller settings)
- `config.yaml` snapshot
- Grasp success / failure (manual visual annotation)

Symmetry argument (rationale for tilt-level choice)

Each gripper is mirror-symmetric across the **xz** plane (perpendicular to the y-axis gripping direction; the two fingers / blades are identical). Combined with the cloth edge being approximately straight (mirror-symmetric about its perpendicular), the two rotations whose axes lie *in* the xz plane — rotation about **x** and rotation about **z** — have +θ ⇔ −θ outcomes under the combined reflection. Only rotation about **y** (the gripping direction itself, perpendicular to the xz plane) breaks this symmetry: both fingers tilt together (fore/aft about the gripping line), and the +y / −y directions correspond to physically different conditions (one tilts the contact toward the cloth body, the other off the edge).

x and z are swept at {0, 2.5°, 5°} (3 levels) by symmetry. y is **physically** asymmetric (the +y and −y tilts are different conditions), but by operator directive (2026-06-04) is swept single-sided at {0, 2.5°, 5°} as well — characterising one tilt direction (into the cloth body) without the −y side. The −y side becomes a follow-up question if the +y data motivates it.

Constitutional / governance notes

- **Article II** (human in the loop): all force-driven descents and grasps run with operator present and e-stop in hand; each session is manually authorized.
- **Article I** (signal first): contact force = Franka `F_ext_z` (Amendment 1 Signal Inventory row 7), bias-corrected per run. A301 FlexiForce calibration is currently blocked by a non-repeatable raw response (suspected TPU 90 A foot spreading load off the active area); F_ext serves as the first-pass primitive until that is resolved.